

SKY BOOK

FLIGHT BOOKING & REVENUE ANALYTICS

End-to-End Cloud-Based Flight Analytics Implementation

Amazon S3 | AWS IAM | AWS Glue | AWS Glue Data Catalog

Amazon Athena | SQL | Athena ODBC | Power BI | Power Query | DAX

PREPARED BY:

Student Name: Jenifer Sathyaseelan

Student ID: AF05251182

Batch Code: ANP-D6641

Course Name: DBA – Data and Business Analysis

PROJECT OBJECTIVE

- This project implements a complete cloud-to-dashboard analytics workflow for a flight booking dataset. The flight data is first placed in Amazon S3 and then made available for structured discovery through AWS Glue. The Glue Data Catalog maintains the metadata required for querying, while Amazon Athena is used to execute SQL directly against the cataloged data.
- After cloud-side preparation and analysis, the flight data is brought into Power BI through the configured Athena connectivity. Power Query is used for data preparation, while DAX measures support the key indicators displayed in the Sky Book dashboard.
- The final report contains two interactive Power BI pages: a Flight Booking Overview and a Revenue & Booking Analytics page. Together, these views cover airline performance, booking volume, passenger information, fares, routes, payment methods, booking status, monthly trends and city-level revenue.

End-to-End Working Flow

Flight Dataset	Amazon S3	AWS Glue	Data Catalog	Amazon Athena	Power BI	Sky Book Dashboard
----------------	-----------	----------	--------------	---------------	----------	--------------------

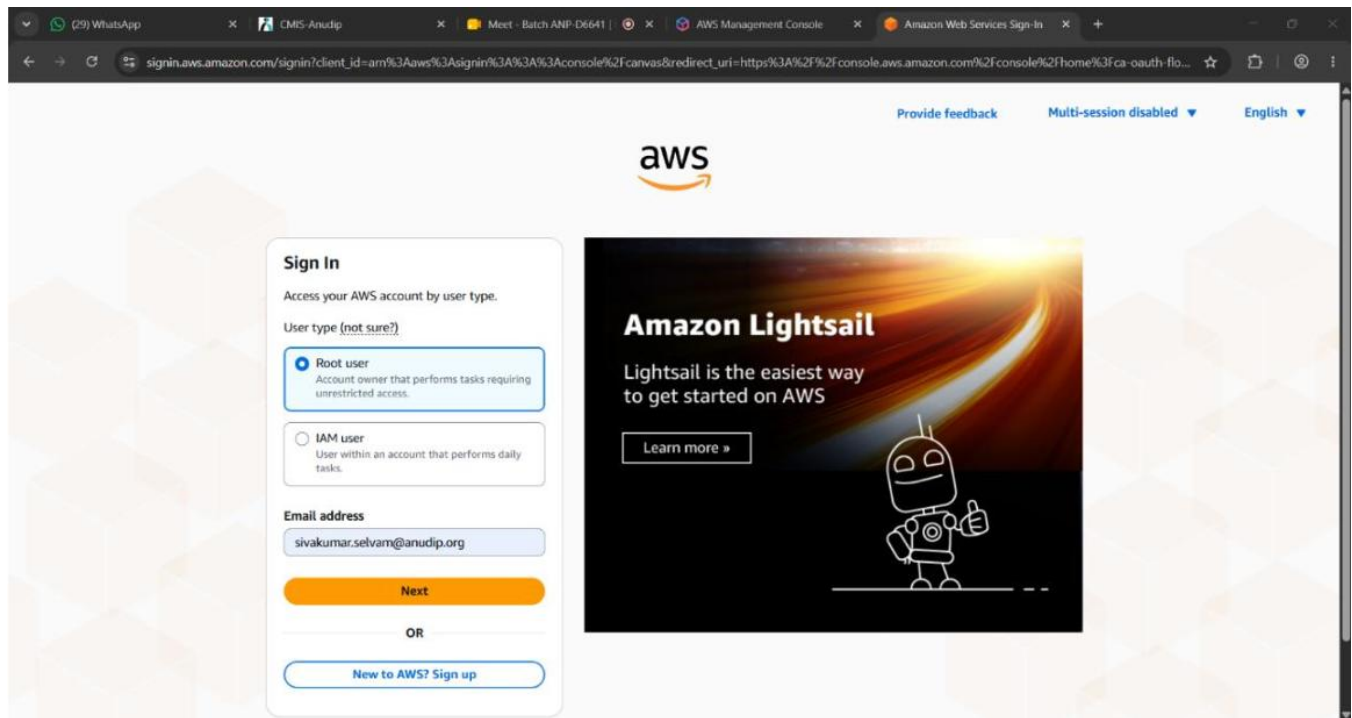
The implementation is documented in the same sequence in which the solution was built, so each stage can be supported with a corresponding screenshot or execution result.

STEP 1: AWS MANAGEMENT CONSOLE ACCESS

The AWS Management Console was used as the starting point for the cloud-based flight analytics workflow. After signing in, the required AWS services were accessed from the console so that storage, permissions, cataloging and query execution could be configured.

Procedure

- Signed in to the AWS Management Console using the project account credentials.
- Completed the required account security checks and accessed the AWS Console home page.
- Used the AWS service search to locate Amazon S3, IAM, AWS Glue and Amazon Athena.
- Confirmed that the services required for the flight-data workflow were available before beginning the configuration.



Result: The AWS environment was ready for creating the storage location and configuring the services required for the flight analytics pipeline.

STEP 2: AMAZON S3 – FLIGHT DATA STORAGE

Amazon S3 was used as the cloud storage layer for the flight dataset. The source file was placed in an S3 location so that AWS Glue could later discover its structure and Amazon Athena could query the cataloged data.

Procedure

- Opened Amazon S3 from the AWS Management Console.
- Created the S3 bucket used for the flight analytics data.
- Created a logical folder for the flight dataset inside the bucket.
- Uploaded the flight booking file to the source location.
- Verified the uploaded file and its S3 path.
- Created or identified the separate S3 location used for Athena query-result files, if configured in the project.

Storage Arrangement

S3 Location	Purpose
Source Bucket / Flight Data Folder	<i>Stores the uploaded flight dataset</i>
Query Result Location	<i>Stores Athena execution results, when configured</i>

The screenshot shows the AWS Management Console interface for the 'ap-south-1' region. The left sidebar contains navigation options for Amazon S3, including Buckets, Files, Access management and security, and Storage management and insights. The main content area displays 'General purpose buckets (20)' with a search bar and a table of buckets. The table has columns for Name, AWS Region, and Creation date. The buckets are listed in a table with the following data:

Name	AWS Region	Creation date
charu-aws-results-bucket	Asia Pacific (Mumbai) ap-south-1	August 28, 2026, 15:15:01 (UTC+05:30)
charu-aws-source-bucket	Asia Pacific (Mumbai) ap-south-1	August 28, 2026, 15:13:36 (UTC+05:30)
devaraj-results-bucket	Asia Pacific (Mumbai) ap-south-1	August 21, 2026, 15:15:09 (UTC+05:30)
devaraj-source-bucket	Asia Pacific (Mumbai) ap-south-1	August 21, 2026, 15:14:23 (UTC+05:30)
dhanu-result	Asia Pacific (Mumbai) ap-south-1	August 24, 2026, 19:55:54 (UTC+05:30)
dhanu-source	Asia Pacific (Mumbai) ap-south-1	August 24, 2026, 19:53:36 (UTC+05:30)
jeni-etl-result	Asia Pacific (Mumbai) ap-south-1	August 28, 2026, 15:57:43 (UTC+05:30)
jeni-etl-source	Asia Pacific (Mumbai) ap-south-1	August 28, 2026, 15:57:00 (UTC+05:30)
kumar-etl-results	Asia Pacific (Mumbai) ap-south-1	August 21, 2026, 15:26:21 (UTC+05:30)
kumar-etl-source	Asia Pacific (Mumbai) ap-south-1	August 21, 2026, 15:23:29 (UTC+05:30)
new-etl-results-bucket	Asia Pacific (Sydney) ap-southeast-2	August 19, 2026, 14:31:50 (UTC+05:30)
new-etl-source-bucket	Asia Pacific (Sydney) ap-southeast-2	August 19, 2026, 14:30:24 (UTC+05:30)
priva-aws-learning	Asia Pacific (Sydney) ap-southeast-2	August 21, 2026, 15:34:42 (UTC+05:30)

STEP 3: AWS IAM – ACCESS CONFIGURATION

AWS IAM was used to control access between the AWS services involved in the workflow. The role assigned to the data-discovery process was configured with permissions needed to read the flight data from S3 and register the discovered metadata in the Glue Data Catalog.

Procedure

- Opened AWS IAM and navigated to Roles.
- Created or selected the IAM role intended for AWS Glue.

- Attached the permissions required for the Glue service to access the project S3 data.
- Reviewed the trust relationship and permissions before saving the role.
- Used the configured role while creating the Glue Crawler.

Permission Objective

Permissions policy summary		
Policy name ↗	Type	Attached as
AmazonS3FullAccess	AWS managed	Permissions policy
AWSGlueServiceRole	AWS managed	Permissions policy

The role should provide only the access required by the workflow. The exact policy names shown in this document should match the policies actually used in the project account; they should not be substituted with unrelated permissions.

STEP 4: AWS GLUE – DATABASE AND CRAWLER

4.1 Create the Glue Database

A database was created in the AWS Glue Data Catalog to provide a metadata container for the flight table discovered from S3.

- Opened AWS Glue and selected Data Catalog → Databases.
- Created a new database for the flight analytics dataset.
- Saved the database and verified that it appeared in the Data Catalog.

Your catalog, more capabilities
Discover, govern, and trust your data in SageMaker Unified Studio with AI-powered exploration, built-in quality scores, and secure sharing across your organization

Databases (6)
Last updated (UTC) August 28, 2026 at 14:40:45

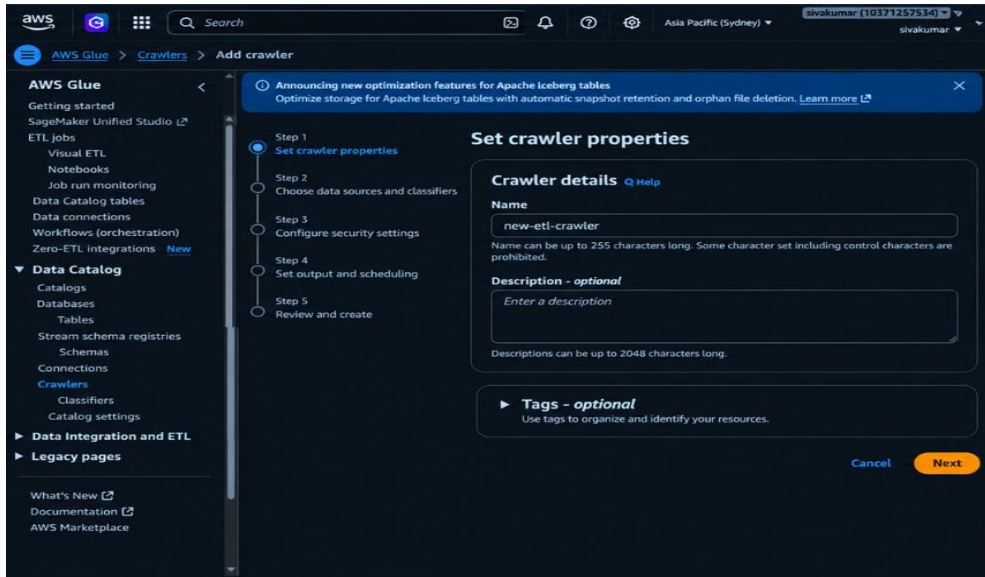
Choose catalog: 103371257534

Name	Description	Location URI	Source catalog	Created on (UTC)
charu-customer-db	-	-	103371257534	August 28, 2026 at 10:18:00
customer-db	-	-	103371257534	August 21, 2026 at 10:03:00
database-db	-	-	103371257534	August 21, 2026 at 10:55:00
dhanu-db	-	-	103371257534	August 24, 2026 at 15:00:00
jeni-customer-db	-	-	103371257534	August 28, 2026 at 10:37:00
praveen-db	-	-	103371257534	August 21, 2026 at 11:09:00

4.2 Create the Glue Crawler

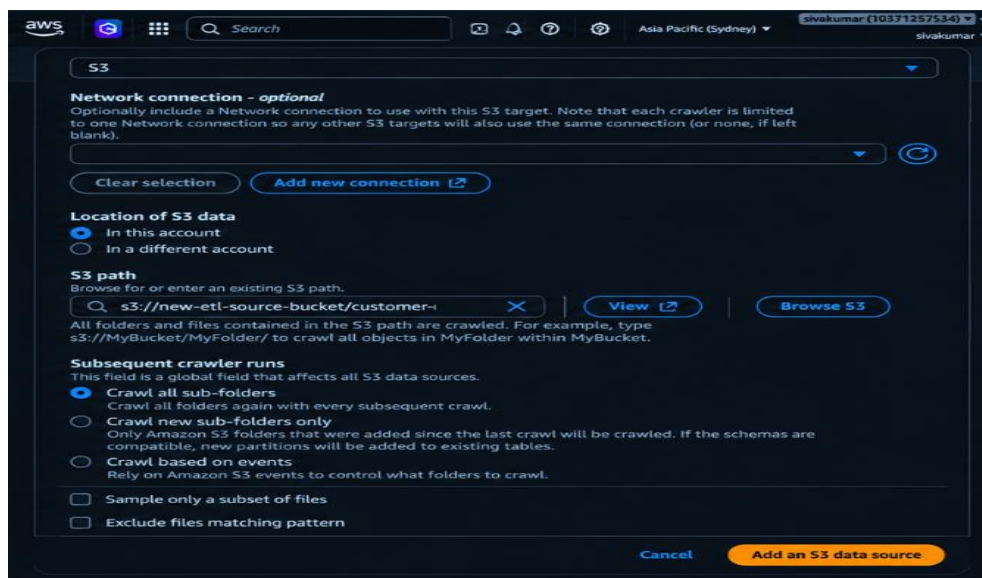
The Glue Crawler was configured to inspect the flight data stored in S3 and automatically infer its structure. The crawler connects the raw storage layer with the cataloged table used later in Athena.

- Opened AWS Glue → Crawlers and selected the option to add a crawler.
- Entered crawler name: new-etl-crawler



On the "Choose data sources and classifiers" screen, selected Not yet for "Is your data already mapped to Glue tables?" since the schema needed to be auto-detected.

- Added the S3 data source:
 - Data source: S3
 - Location of S3 data: In this account
 - S3 path: s3://new-etl-source-bucket/customer-db/
 - Subsequent crawler runs: Crawl all sub-folders



- Selected the IAM role created in Step 3 (new-etl-role) under "Configure security settings" so the crawler has permission to read S3 and write to the Glue Data Catalog.
- Reviewed all settings on the "Review and create" screen and created the crawler
- IAM role: new-etl-role

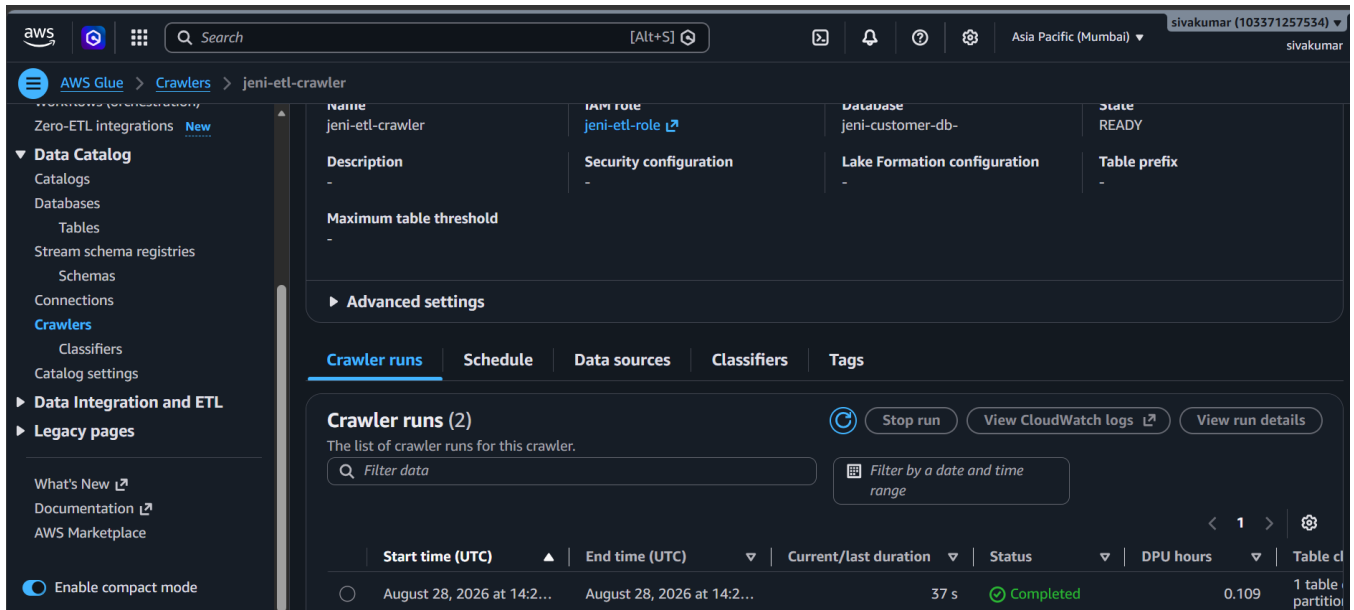
The screenshot shows the AWS Glue Crawlers console. The left sidebar contains navigation links for Data Catalog, Data Integration and ETL, and Legacy pages. The main panel displays a table of crawlers with columns for Name, State, Schedule, Last run, Last run time, Log, and Table change. Five crawlers are listed, all with a 'Ready' state and 'Succeeded' last run status.

Name	State	Schedule	Last run	Last run time	Log	Table change...
charu-etl-cra...	Ready		Succeeded	August 28, 20...	View log	1 created
dhanu-crawler	Ready		Succeeded	August 24, 20...	View log	1 created
jeni-etl-crawler	Ready		Succeeded	August 28, 20...	View log	1 created
kumar-etl-cra...	Ready		Succeeded	August 21, 20...	View log	1 created
praveen-crawl...	Ready		Succeeded	August 21, 20...	View log	1 created

4.3 Run the Crawler and Verify the Table

After the crawler configuration was completed, it was executed manually. The crawler inspected the S3 file, identified the available fields and created the corresponding table in the Glue Data Catalog.

- Started the configured Glue Crawler manually.
- Monitored the crawler until the execution status changed to Completed.
- Opened the Data Catalog and selected the database created for the project.
- Verified that the flight table was created successfully.
- Reviewed the detected column names and data types.
- Confirmed that the cataloged table was ready for Amazon Athena queries.

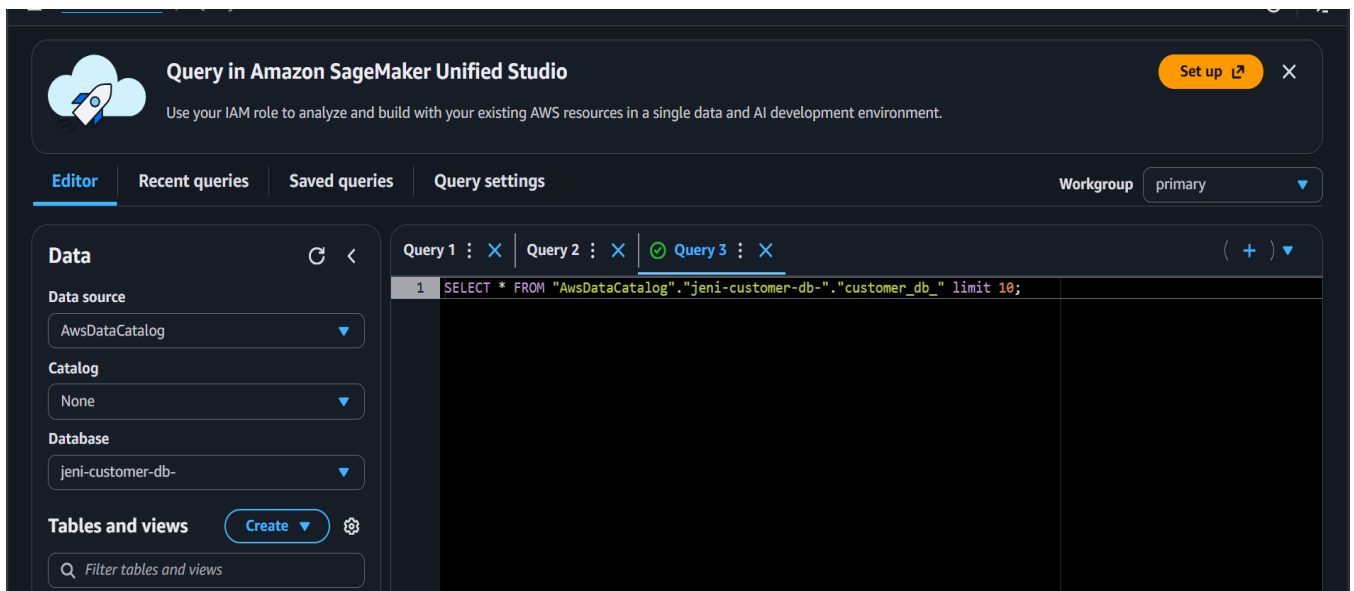


STEP 5: AMAZON ATHENA – SQL ANALYSIS

Amazon Athena was used as the serverless SQL layer for analysing the cataloged flight table. The database and table created by the Glue Crawler were selected before running the analytical queries.

Athena Preparation

- Opened Amazon Athena from the AWS Management Console.
- Selected the Glue Data Catalog and the database containing the flight table.
- Selected the cataloged flight table for analysis.
- Configured the S3 query-result location used by Athena.
- Executed SQL statements for booking, airline, route, fare, payment, status, city and time-based analysis.



SQL Analysis Groups

- Part 1 – Basic record checks and booking metrics
- Part 2 – Airline, route and booking-status aggregation
- Part 3 – Fare, age, payment and city-level analysis
- Part 4 – Monthly trends and top-value records

AWS SQL Results – Part 1: Core Flight Metrics

The following queries were designed around the fields represented in the Sky Book dashboard. Before execution, replace the catalog, database and table identifiers with the exact names generated in the user's AWS environment.

1. Total Booking Records

```
SELECT COUNT(*) AS total_bookings
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

2. Total Revenue

```
SELECT SUM("fare") AS total_revenue
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

3. Average Fare

```
SELECT AVG("fare") AS average_fare
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

4. Minimum Fare

```
SELECT MIN("fare") AS minimum_fare
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

5. Maximum Fare

```
SELECT MAX("fare") AS maximum_fare
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

AWS SQL Results – Part 2: Airline, Route & Status

6. Airline-wise Booking Count

```
SELECT "airline", COUNT(*) AS booking_count
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "airline"
ORDER BY booking_count DESC;
```

7. Airline-wise Revenue

```
SELECT "airline", SUM("fare") AS total_revenue
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "airline"
ORDER BY total_revenue DESC;
```

8. Route-wise Booking Count

```
SELECT "route", COUNT(*) AS booking_count
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "route"
ORDER BY booking_count DESC;
```

9. Booking Status Distribution

```
SELECT "bookingstatus", COUNT(*) AS booking_count
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "bookingstatus"
ORDER BY booking_count DESC;
```

10. Revenue by Booking Status

```
SELECT "bookingstatus", SUM("fare") AS total_revenue
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "bookingstatus"
ORDER BY total_revenue DESC;
```

AWS SQL Results – Part 3: Passenger, Payment & City Analysis

11. Average Passenger Age

```
SELECT AVG("age") AS average_age
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

12. Age Range in the Dataset

```
SELECT MIN("age") AS minimum_age,
       MAX("age") AS maximum_age
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE";
```

13. Payment Method Distribution

```
SELECT "payment_method", COUNT(*) AS booking_count
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
```

```
GROUP BY "payment_method"
ORDER BY booking_count DESC;
```

14. Payment Method Revenue

```
SELECT "payment_method", SUM("fare") AS total_revenue
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "payment_method"
ORDER BY total_revenue DESC;
```

15. Departure City Booking Count

```
SELECT "departure_city", COUNT(*) AS booking_count
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "departure_city"
ORDER BY booking_count DESC;
```

AWS SQL Results - Part 4: Trend & Ranking Analysis

16. City-wise Revenue

```
SELECT "departure_city", SUM("fare") AS total_revenue
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
GROUP BY "departure_city"
ORDER BY total_revenue DESC;
```

17. Top 10 Highest Fare Bookings

```
SELECT *
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
ORDER BY "fare" DESC
LIMIT 10;
```

18. Confirmed Booking Records

```
SELECT COUNT(*) AS confirmed_bookings
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
WHERE "bookingstatus" = 'Confirmed';
```

19. Cancelled Booking Records

```
SELECT COUNT(*) AS cancelled_bookings
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
WHERE "bookingstatus" = 'Cancelled';
```

20. Completed Booking Records

```
SELECT COUNT(*) AS completed_bookings
FROM "AwsDataCatalog"."YOUR_DATABASE"."YOUR_FLIGHT_TABLE"
WHERE "bookingstatus" = 'Completed';
```

STEP 6: ATHENA CONNECTIVITY AND POWER BI DATA ACCESS

Once the AWS-side dataset and SQL layer were ready, the cataloged flight data was made available to Power BI through the configured Athena connectivity. This stage forms the bridge between cloud-side processing and dashboard development.

6.1 Configure Athena Connectivity

- Installed or used the configured Amazon Athena ODBC connectivity on the Windows system.
- Created or selected the required Data Source Name (DSN).
- Verified that the connection could access the cataloged flight table.

6.2 Connect from Power BI

- Opened Power BI Desktop.
- Selected Get Data and chose the configured ODBC/Athena connection.
- Authenticated the connection and selected the flight database/table.
- Loaded the required data into Power BI.
- Opened Power Query to validate and prepare the fields for reporting.

STEP 7: POWER QUERY, DAX AND DASHBOARD DEVELOPMENT

7.1 Power Query Preparation

- Checked the data types of flight date, fare, age and categorical fields.
- Standardised field values where required for consistent filtering and grouping.
- Validated the booking status, airline, route, city and payment-method fields.
- Prepared the date field for monthly trend analysis.
- Applied the transformation steps and loaded the prepared data into the Power BI model.

7.2 DAX Measures Used for the Dashboard

Total Bookings = COUNTROWS(FlightData)
Total Revenue = SUM(FlightData[fare])
Average Fare = AVERAGE(FlightData[fare])
Average Age = AVERAGE(FlightData[age])
Confirmed Bookings = CALCULATE(COUNTROWS(FlightData), FlightData[bookingstatus] = "Confirmed")
Cancelled Bookings = CALCULATE(COUNTROWS(FlightData), FlightData[bookingstatus] = "Cancelled")
Completed Bookings = CALCULATE(COUNTROWS(FlightData), FlightData[bookingstatus] = "Completed")

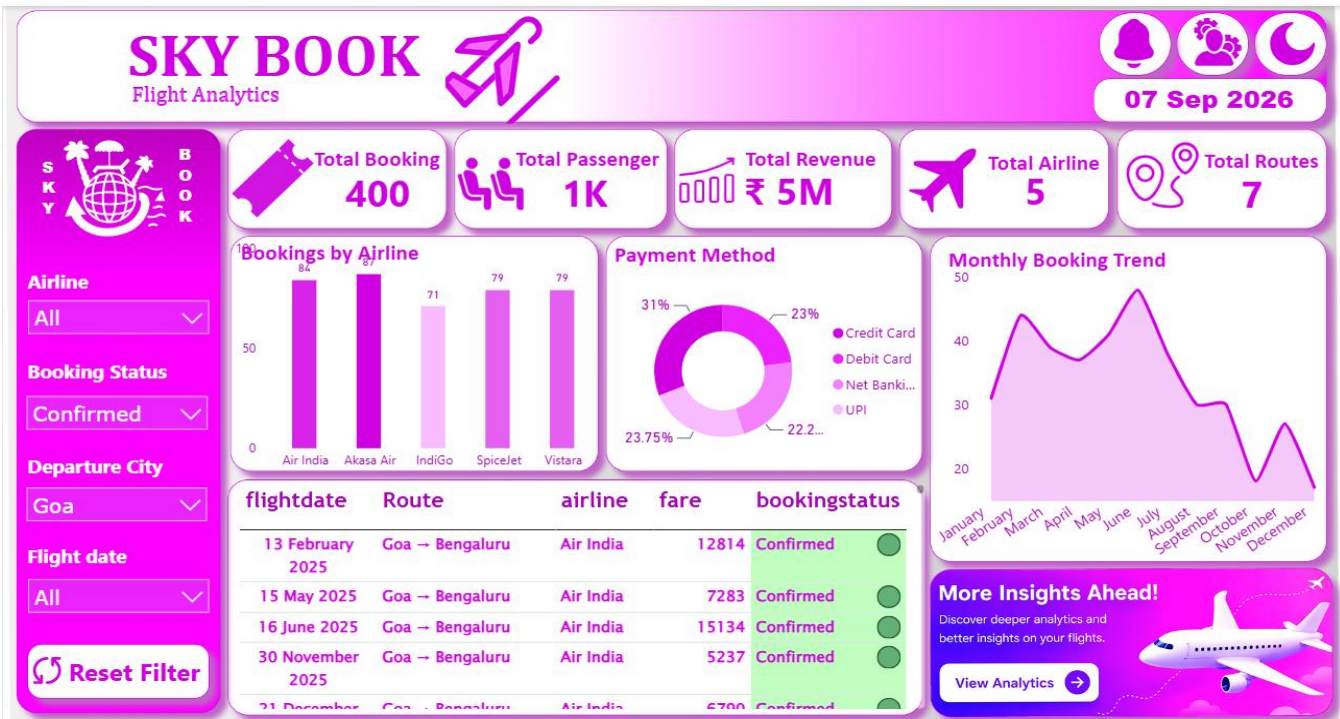
7.3 Dashboard Construction

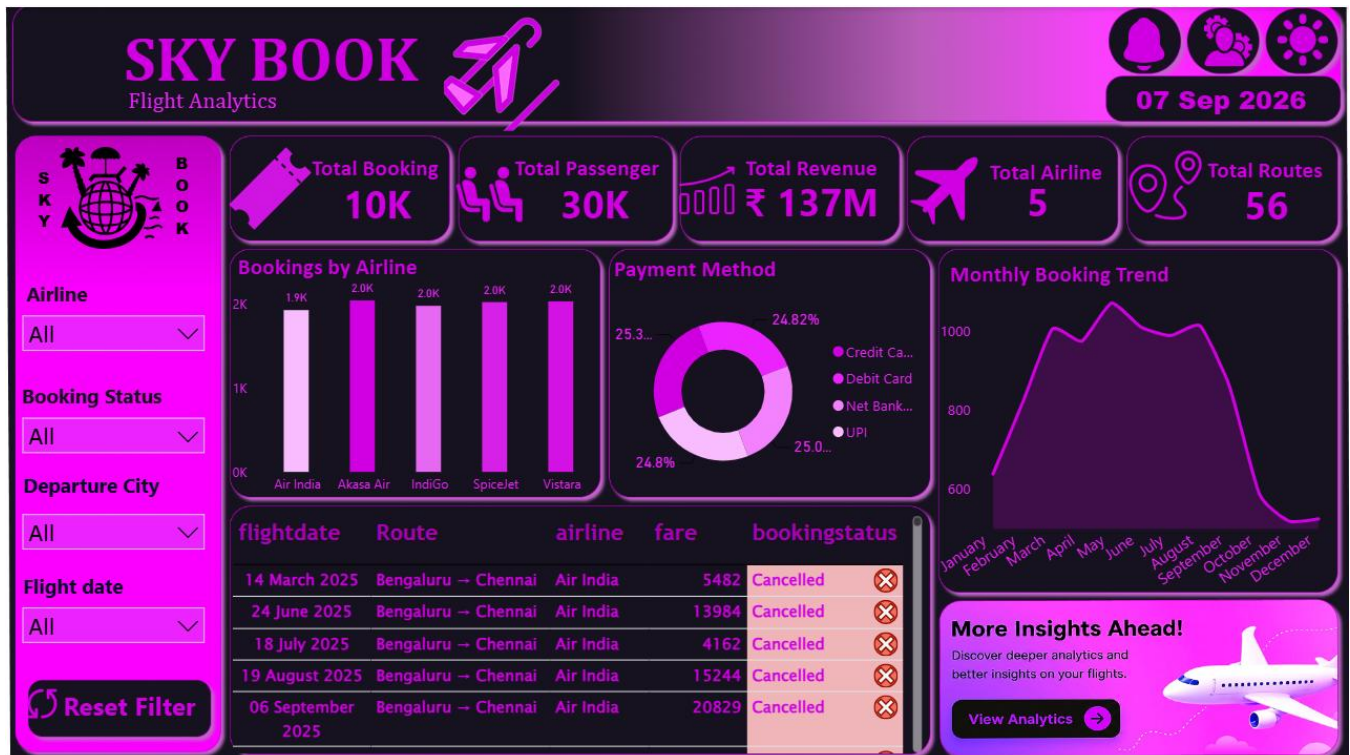
- The prepared model was used to build two Power BI pages.
- The first page focuses on the overall flight booking picture
- The second page concentrates on revenue and booking-performance analysis.

STEP 8: SKY BOOK DASHBOARD & SERVICES USED

8.1 Flight Booking Overview

The first dashboard page combines the main business indicators with airline, payment, monthly and booking-level views. The interactive filters allow the report user to narrow the analysis by airline, booking status, departure city and flight date.

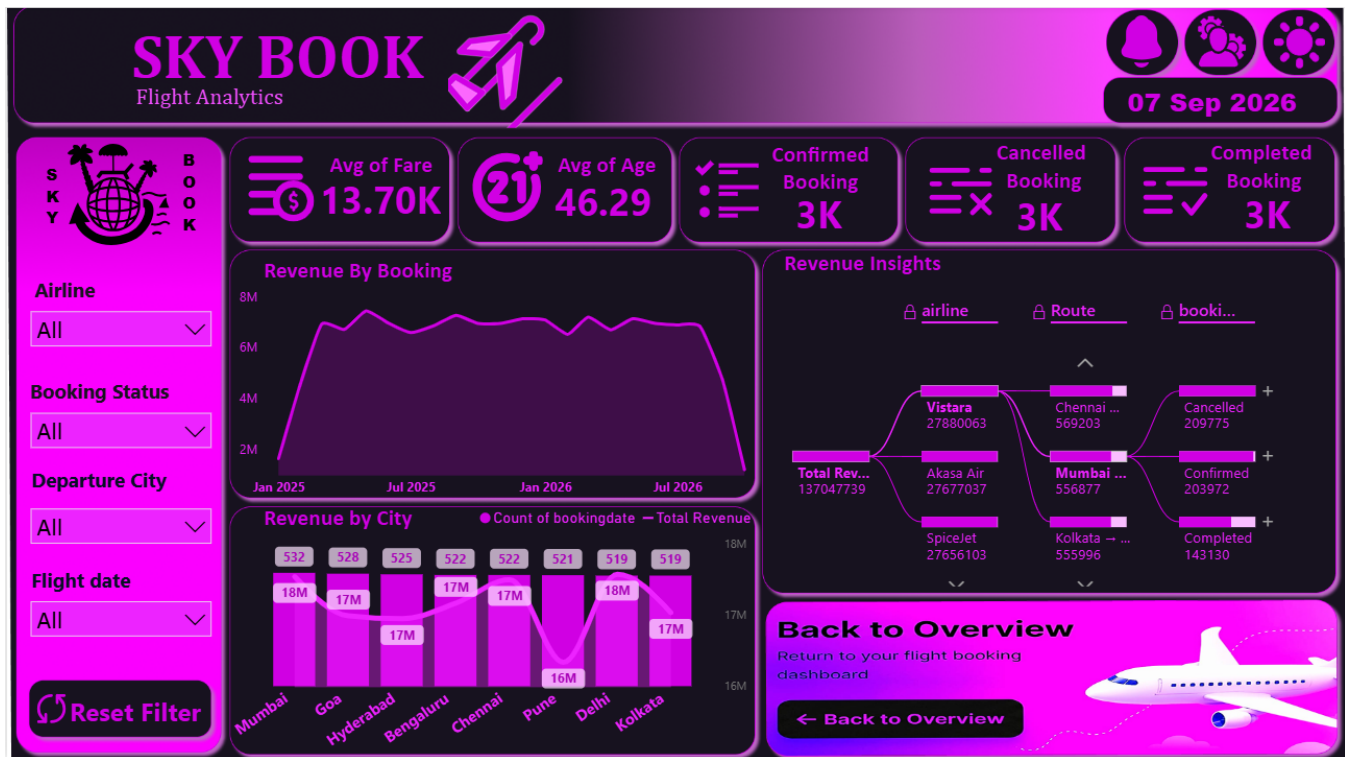




8.2 Revenue & Booking Analytics

The second dashboard page provides a deeper financial and operational view using average fare, passenger age, booking-status indicators, revenue trends, city-level revenue and a revenue-driver analysis.





Services Used

Service / Tool	Role in the Project
Amazon S3	Cloud storage for the flight dataset
AWS IAM	Access and role management
AWS Glue	Crawler-based schema discovery
Glue Data Catalog	Metadata and table registration
Amazon Athena	Serverless SQL analysis
SQL	Flight booking and revenue analysis
Athena ODBC	Connectivity between Athena and Power BI
Power Query	Data preparation and transformation
DAX	Dynamic KPI and business calculations
Power BI	Interactive dashboard development